

Parameter Limits (BIS IS 10500:2012 + (PCB))

Parameter	Acceptable limit (S_n)	ideal V_{ideal}
1) PH	6.5-8.5	7.0
2) Turbidity	≤ 1 NTU	0
3) TDS	≤ 500 mg/L	0
4) EC	≤ 225 μ S/cm	0
5) Dissolved Oxygen	≥ 5 mg/L	14.6
6) Temperature	advisory	25

Safe/ Unsafe Classification (or per reading)

For each parameter V :

- ① PH
- $$\left\{ \begin{array}{l} \text{Critical} \\ \text{Unsafe} \\ \text{Safe} \end{array} \right. \left\{ \begin{array}{l} V < 5.5 \text{ or } V > 9.5 \\ V < 6.5 \text{ or } V > 8.5 \\ 6.5 \leq V \leq 8.5 \end{array} \right.$$
- ② Turbidity, TDS, Iron, EC (lower is better.)
- $$\left\{ \begin{array}{l} \text{Critical} \\ \text{Unsafe} \\ \text{Safe} \end{array} \right. \left\{ \begin{array}{l} V > S_{perm} \text{ (or } V > 2S_n) \\ V > S_n \\ V \leq S_n \end{array} \right.$$
- ③ Dissolved Oxygen
- $$\left\{ \begin{array}{l} \text{critical} \\ \text{safe} \\ \text{unsafe} \end{array} \right. \left\{ \begin{array}{l} V < 4 \\ V \geq 5 \\ V < 5 \end{array} \right.$$

Water Quality Index (WQI) - weighted arithmetic method

Step 1 : Unit weight

$$K = \frac{1}{\sum_{n=1}^N \frac{1}{S_n}}, \quad w_n = \frac{K}{S_n} \quad (w_n \text{ sums to } 1)$$

Step 2 : Quality rating sub-Index

$$q_n = 100 \times \left| \frac{V_n - V_{\text{ideal}}}{S_n - V_{\text{ideal}}} \right|$$

Step 3 : Overall Index

$$WQI = \frac{\sum_{n=1}^N (q_n \cdot w_n)}{\sum_{n=1}^N w_n}$$

<u>Rating</u> :	WQI range	Rating
	0-25	Excellent
	26-50	Good
	51-75	Poor
	76-100	Very Poor
	>100	Unfit for drinking

Compliance percentage (Over a period)

$$\text{compliance \%} = \frac{N_{\text{within limit}}}{N_{\text{total}}} \times 100$$

Assessment :

$\geq 95\%$ $\rightarrow$ satisfactory

80-95% $\rightarrow$ needs attention

$< 80\%$ $\rightarrow$ Non-compliant

Purification / Verification loop

Removal efficiency $\eta = \frac{C_{in} - C_{out}}{C_{in}} \times 100\%$

↳ Apply this to Turbidity, TDS etc.

Retreatment decision

{ Pass : all outlet parameters within limit }
{ Re-treat : any parameter still unsafe / critical }

Priority for treatment (targeting)

$$\text{Priority} = 3 \times N_{\text{critical}} + 2 \times N_{\text{unsafe}} + 1 \times N_{\text{marginal}}$$

Where,

N_{critical} = number of parameters that are critical

N_{unsafe} = number of parameters that are unsafe

N_{marginal} = number of parameters that are Marginal

Rule :

Status of source	Priority score	Meaning
Critical	3	Treat first
Unsafe	2	Treat next
Marginal	1	Treat later
Safe	0	Do not treat

Lets Consider an Example

Suppose,

$$\text{pH} = 6.53$$

$$\text{Turbidity} = 48.29 \text{ NTU}$$

$$\text{TDS} = 704 \text{ mg/L}$$

$$\text{EC} = 1092 \text{ MS/cm}$$

Then,

- pH $\rightarrow$ safe (inside 6.5 - 8.5)
- Turbidity $\rightarrow$ Critical ($\gg 1 \text{ NTU}$)
- TDS $\rightarrow$ Unsafe (> 500)
- EC $\rightarrow$ Safe (≤ 2250)
- Node Verdict = Critical (worst wins)
- Trigger targeted purification $\rightarrow$ re-measure $\rightarrow$ check η and Safe again